DAILY PRACTICE CALENDAR

Grade 8 Math • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **California Common Core standards for Grade 8**. Students who want an extra challenge can add **Geometry Competition Math**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Exponent Rules
2	Aug 23–Aug 27	Scientific Notation
3	Aug 30–Sep 3	Square Roots & Cube Roots
4	Sep 6–Sep 10	Rational & Irrational Numbers
5	Sep 13–Sep 17	Multi-Step Equations
6	Sep 20–Sep 24	Variables on Both Sides
7	Sep 27–Oct 1	One, None or Infinitely Many Solutions
8	Oct 4–Oct 8	Slope & Proportional Relationships
9	Oct 11–Oct 15	Slope-Intercept Form
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	What Is a Function?
12	Nov 1–Nov 5	Linear & Nonlinear Functions
13	Nov 8–Nov 12	Comparing & Modeling Functions
14	Nov 15–Nov 19	Systems of Linear Equations
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Systems Word Problems
16	Dec 6–Dec 10	Translations, Reflections & Rotations
17	Dec 13–Dec 17	Dilations
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Congruence & Similarity
19	Jan 10–Jan 14	Angles: Parallel Lines & Triangles
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Angle-Angle Similarity
22	Jan 31–Feb 4	The Pythagorean Theorem
23	Feb 7–Feb 11	The Converse & Applications
24	Feb 14–Feb 18	Distance on the Coordinate Plane
25	Feb 21–Feb 25	Volume of Cylinders & Cones
26	Feb 28–Mar 3	Volume of Spheres
27	Mar 6–Mar 10	Scatter Plots
28	Mar 13–Mar 17	Lines of Best Fit
29	Mar 20–Mar 24	Using Linear Models
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Two-Way Tables
31	Apr 10–Apr 14	Relative Frequency & Association
32	Apr 17–Apr 21	Year-End Review: Numbers, Equations & Functions
33	Apr 24–Apr 28	Year-End Review & Practice Test
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Exponents & Real Numbers

Exponent Rules

Core: Same base: add exponents to multiply, subtract to divide; $a^0 = 1$ and $a^{-n} = \frac{1}{a^n}$.

Mon Aug 16

Product rule

TRY Write $3^4 \times 3^2$ as a single power of 3.

Tue Aug 17

Quotient rule

TRY Compute $5^7 \div 5^5$.

Wed Aug 18

Zero & negative exponents

TRY Find 2^{-3} and 7^0.

Thu Aug 19

Mixing the rules

TRY Compute $3^2 \times 3^{-5}$.

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Exponents & Real Numbers

Scientific Notation

Core: Write numbers as $a \times 10^n$ with $1 \le a < 10$.

Mon Aug 23

Large numbers

TRY Write 93,000,000 in scientific notation.

Tue Aug 24

Small numbers

TRY Write 0.00042 in scientific notation.

Wed Aug 25

Multiplying

TRY Compute $(3 \times 10^4)(2 \times 10^5)$.

Thu Aug 26

How many times larger?

TRY How many times larger is 6×10^8 than 2×10^5?

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Exponents & Real Numbers

Square Roots & Cube Roots

Core: Square roots undo squaring; cube roots undo cubing.

Mon Aug 30

Perfect squares

TRY Find $\sqrt{144}$ and $\sqrt{\frac{49}{64}}$.

Tue Aug 31

Cube roots

TRY Find $\sqrt[3]{125}$ and $\sqrt[3]{-8}$.

Wed Sep 1

Solving $x^2 = p$

TRY Solve $x^2 = 81$.

Thu Sep 2

Solving $x^3 = p$

TRY A cube has volume 343 cm^3. How long is each edge?

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Exponents & Real Numbers

Rational & Irrational Numbers

Core: Irrational numbers never end and never repeat; estimate them with nearby squares.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Rational or irrational?

TRY Classify $\sqrt{16}$, $\sqrt{7}$, 0.25 and π.

Tue Sep 7

Repeating decimals

TRY Write $0.\overline{4}$ as a fraction.

Wed Sep 8

Estimating roots

TRY Between which two integers is $\sqrt{50}$? Which is it closer to?

Thu Sep 9

Ordering real numbers

TRY Order from least to greatest: $\sqrt{10}$, 3.1, π, $3\frac{1}{4}$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Linear Equations & Slope

Multi-Step Equations **Core:** Distribute, combine like terms, then undo the operations.

Mon Sep 13
Distributing
TRY Solve $3(2x - 4) = 18$.

Tue Sep 14
Combining like terms
TRY Solve $5x - 2x + 7 = 19$.

Wed Sep 15
Fractions
TRY Solve $\frac{x+3}{4} = 5$.

Thu Sep 16
Decimals
TRY Solve $0.4x + 1.2 = 3.6$.

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Linear Equations & Slope

Variables on Both Sides **Core:** Collect the variable terms on one side and the numbers on the other.

Mon Sep 20
The basics
TRY Solve $7x + 3 = 4x + 18$.

Tue Sep 21
With distributing
TRY Solve $2(x + 5) = 5x - 8$.

Wed Sep 22
Fractions
TRY Solve $\frac{x}{2} + 3 = \frac{x}{3} + 5$.

Thu Sep 23
Word problems
TRY Gym A charges $30 plus $5 a visit. Gym B charges $10 a visit. After how many visits do they cost the same?

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Linear Equations & Slope

One, None or Infinitely Many Solutions **Core:** $x = a$: one solution. $a = a$: infinitely many. $a = b$ with $a \neq b$: none.

Mon Sep 27
One solution
TRY Solve $4x - 1 = 2x + 9$.

Tue Sep 28
No solution
TRY How many solutions does $3(x + 2) = 3x + 5$ have?

Wed Sep 29
Infinitely many
TRY How many solutions does $2(x - 3) + x = 3x - 6$ have?

Thu Sep 30
Make it happen
TRY What number in the box gives $5x + 2 = 5x + \square$ infinitely many solutions?

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Linear Equations & Slope

Slope & Proportional Relationships **Core:** Slope $= \frac{\text{rise}}{\text{run}}$; the graph of $y = kx$ is a line through the origin with slope k.

Mon Oct 4
Slope from two points
TRY Find the slope of the line through $(1, 3)$ and $(4, 12)$.

Tue Oct 5
Comparing rates
TRY Car A travels $y = 55x$ miles in x hours. Car B goes 180 miles in 3 hours. Which is faster?

Wed Oct 6
Negative slope
TRY Find the slope of the line through $(-2, 5)$ and $(2, -3)$.

Thu Oct 7
Why slope is constant
TRY Why is the slope the same between any two points on a line?

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Linear Equations & Slope

Slope-Intercept Form — **Core:** In $y = mx + b$, m is the slope and b is the y-intercept.

Mon Oct 11
Reading m and b
TRY Find the slope and y-intercept of $y = -3x + 7$.

Tue Oct 12
Slope & intercept
TRY Write the line with slope 2 that passes through $(0, -5)$.

Wed Oct 13
Two points
TRY Write the line through $(1, 4)$ and $(3, 10)$.

Thu Oct 14
Real-world lines
TRY A plumber charges a $40 fee plus $25 an hour. Write the cost y for x hours.

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Linear Equations & Slope

Review & Checkpoint 1 — **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: exponents
TRY Find 4^{-2}.

Tue Oct 19
Review: scientific notation
TRY Compute $(8 \times 10^6) \div (2 \times 10^2)$.

Wed Oct 20
Review: equations
TRY Solve $6x - 4 = 2x + 12$.

Thu Oct 21
Review: slope
TRY Find the slope of the line through $(0, 2)$ and $(5, 12)$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Functions & Systems

What Is a Function? — **Core:** A function gives exactly one output for each input.

Mon Oct 25
Function or not?
TRY Is $\{(1, 2), (2, 4), (1, 5)\}$ a function?

Tue Oct 26
Input–output rules
TRY The rule is $y = 2x + 1$. What is the output for input 4?

Wed Oct 27
The vertical line test
TRY Is a circle the graph of a function?

Thu Oct 28
Function notation
TRY If $f(x) = x^2 - 1$, find $f(3)$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Functions & Systems

Linear & Nonlinear Functions — **Core:** A linear function changes by the same amount each step; its graph is a line.

Mon Nov 1
From tables
TRY A table has x: 0, 1, 2, 3 and y: 1, 3, 9, 27. Is it linear?

Tue Nov 2
From equations
TRY Is $y = x^2$ linear? Is $y = 4 - x$?

Wed Nov 3
Rate of change
TRY A table has x: 2, 4, 6 and y: 7, 11, 15. Find the rate of change.

Thu Nov 4
Describing graphs
TRY A graph of height over time rises steeply, then levels off. Describe it in words.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Functions & Systems

Comparing & Modeling Functions

Core: Compare rates of change and starting values; write a model from a story.

Mon Nov 8
Comparing rates
TRY Function f is $y = 3x + 2$. Function g passes through $(0, 1)$ and $(2, 9)$. Which has the greater rate of change?

Tue Nov 9
Comparing starting values
TRY For the same functions, which has the greater y-intercept?

Wed Nov 10
Writing a model
TRY A pool holds 500 gallons and drains 20 gallons a minute. Write the gallons y left after x minutes.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Using a model
TRY Using Wednesday's model, when will the pool be empty?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Functions & Systems

Systems of Linear Equations

Core: The solution of a system is the point where the lines meet.

Mon Nov 15
Checking a solution
TRY Is $(2, 3)$ a solution of $y = x + 1$ and $y = 2x - 1$?

Tue Nov 16
Substitution
TRY Solve $y = 3x$ and $x + y = 20$.

Wed Nov 17
Elimination
TRY Solve $x + y = 10$ and $x - y = 4$.

Thu Nov 18
No solution
TRY How many solutions do $y = 2x + 1$ and $y = 2x - 3$ have?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Functions & Systems

Systems Word Problems

Core: Two unknowns need two equations.

Mon Nov 29
Tickets
TRY Adult tickets cost \$12 and child tickets \$7. 30 tickets sold for \$285. How many of each were sold?

Tue Nov 30
Comparing plans
TRY Plan A costs \$20 plus \$0.05 a minute; Plan B costs \$0.15 a minute. At how many minutes do they cost the same?

Wed Dec 1
Sum & difference
TRY Two numbers add to 56 and differ by 14. Find them.

Thu Dec 2
Coins
TRY 15 nickels and dimes are worth \$1.10. How many of each are there?

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Transformations, Congruence & Similarity

Translations, Reflections & Rotations

Core: Slides, flips and turns keep size and shape the same.

Mon Dec 6
Translations
TRY Translate $(3, -2)$ 4 units left and 5 units up.

Tue Dec 7
Reflections
TRY Reflect $(5, 2)$ over the x-axis.

Wed Dec 8
Rotations
TRY Rotate $(4, 1)$ by $90°$ counterclockwise about the origin.

Thu Dec 9
What stays the same
TRY After a rotation, are the side lengths and angles the same?

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Transformations, Congruence & Similarity

Dilations **Core:** A dilation from the origin multiplies each coordinate by the scale factor k.

Mon Dec 13
Enlarging
TRY Dilate $(2, -3)$ by a scale factor of 3 from the origin.

Tue Dec 14
Shrinking
TRY Dilate $(8, 4)$ by a scale factor of $\frac{1}{2}$ from the origin.

Wed Dec 15
Finding the scale factor
TRY A side of length 6 becomes 15 after a dilation. Find the scale factor.

Thu Dec 16
What stays the same
TRY In a dilation, which stay the same: side lengths or angles?

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Transformations, Congruence & Similarity

Congruence & Similarity **Core:** Congruent: slides, flips and turns only. Similar: add a dilation.

Mon Jan 3
Congruent or similar?
TRY A shape is reflected and then dilated by a factor of 2. Is the image congruent or similar?

Tue Jan 4
Describing moves
TRY Name one transformation that takes $(1, 1)$ to $(-1, -1)$.

Wed Jan 5
Corresponding sides
TRY A 3-4-5 triangle is similar to one with sides 9 and 12. Find the third side.

Thu Jan 6
Missing lengths
TRY A 4×6 rectangle is similar to a rectangle with short side 10. Find its long side.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Transformations, Congruence & Similarity

Angles: Parallel Lines & Triangles **Core:** With parallel lines, corresponding and alternate angles are equal.

Mon Jan 10
Corresponding angles
TRY Parallel lines are cut by a transversal, and one angle is $65°$. Find its corresponding angle.

Tue Jan 11
Same-side interior angles
TRY One same-side interior angle is $110°$. Find the other.

Wed Jan 12
Triangle angle sum
TRY A triangle has angles $x°$, $2x°$ and $3x°$. Find all three.

Thu Jan 13
Exterior angles
TRY The two remote interior angles are $48°$ and $67°$. Find the exterior angle.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Transformations, Congruence & Similarity

Mid-Year Review & Checkpoint **Core:** Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: square roots
TRY Between which two integers is $\sqrt{30}$?

Tue Jan 18
Review: equations
TRY Solve $4(x - 1) = 2x + 6$.

Wed Jan 19
Review: functions
TRY Find the rate of change of $y = -4x + 9$.

Thu Jan 20
Review: systems
TRY Solve $y = x + 2$ and $y = -x + 6$.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 4: Transformations, Congruence & Similarity

Angle-Angle Similarity

Core: Two pairs of equal angles are enough to make two triangles similar.

Mon Jan 24
AA similarity
TRY One triangle has angles $40°$ and $60°$; another has $60°$ and $80°$. Are they similar?

Tue Jan 25
Shadows
TRY A 1.8 m person casts a 2.4 m shadow. A tree casts a 12 m shadow. How tall is the tree?

Wed Jan 26
Nested triangles
TRY In $\triangle ABC$, $DE \parallel BC$ with D on AB. $AD = 4$, $AB = 10$, $BC = 15$. Find DE.

Thu Jan 27
Not similar
TRY One triangle has angles $50°$ and $70°$; another has $50°$ and $80°$. Are they similar?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: The Pythagorean Theorem & Volume

The Pythagorean Theorem

Core: In a right triangle, $a^2 + b^2 = c^2$.

Mon Jan 31
Finding the hypotenuse
TRY A right triangle has legs 6 and 8. Find the hypotenuse.

Tue Feb 1
Finding a leg
TRY The hypotenuse is 17 and one leg is 8. Find the other leg.

Wed Feb 2
Ladders
TRY A 13-ft ladder leans on a wall with its foot 5 ft from the wall. How high does it reach?

Thu Feb 3
Irrational answers
TRY A right triangle has legs 2 and 3. Find the hypotenuse.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: The Pythagorean Theorem & Volume

The Converse & Applications

Core: If $a^2 + b^2 = c^2$, the triangle is a right triangle.

Mon Feb 7
The converse
TRY Is a triangle with sides 7, 24 and 25 a right triangle?

Tue Feb 8
Not a right triangle
TRY Is a triangle with sides 6, 7 and 9 a right triangle?

Wed Feb 9
Screen sizes
TRY A TV screen is 32 in. wide and 24 in. tall. How long is its diagonal?

Thu Feb 10
Diagonals in 3D
TRY What is the longest straight rod that fits inside a $3 \times 4 \times 12$ box?

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: The Pythagorean Theorem & Volume

Distance on the Coordinate Plane

Core: Draw a right triangle, then use the Pythagorean theorem.

Mon Feb 14
Distance
TRY Find the distance between $(1, 2)$ and $(4, 6)$.

Tue Feb 15
Across quadrants
TRY Find the distance between $(-3, -1)$ and $(3, 7)$.

Wed Feb 16
Perimeter
TRY Find the perimeter of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(4, 3)$.

Thu Feb 17
Longest side
TRY A triangle has vertices $(0, 0)$, $(5, 0)$ and $(0, 12)$. Find its longest side.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: The Pythagorean Theorem & Volume

Volume of Cylinders & Cones — **Core:** Cylinder: $V = \pi r^2 h$. Cone: $V = \frac{1}{3}\pi r^2 h$.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Cylinders

TRY Find the volume of a cylinder with radius 3 and height 10.

Tue Feb 22

Cones

TRY Find the volume of a cone with radius 3 and height 10.

Wed Feb 23

Missing height

TRY A cylinder has volume 100π and radius 5. Find its height.

Thu Feb 24

Cones vs. cylinders

TRY A cone and a cylinder have the same base and height. How many cones of water fill the cylinder?

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: The Pythagorean Theorem & Volume

Volume of Spheres — **Core:** $V = \frac{4}{3}\pi r^3$.

Mon Feb 28

Spheres

TRY Find the volume of a sphere with radius 3.

Tue Feb 29

Hemispheres

TRY Find the volume of a hemisphere with radius 6.

Wed Mar 1

A ball in a box

TRY A ball with diameter 10 fits snugly in a cube-shaped box with edge 10. How much space is left?

Thu Mar 2

Ice cream cones

TRY A cone with radius 2 and height 6 is topped by a hemisphere of radius 2. Find the total volume.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Data & Statistics

Scatter Plots — **Core:** Look for direction, form, strength and outliers.

Mon Mar 6

Positive association

TRY What kind of association do hours studied and test scores usually show?

Tue Mar 7

Negative association

TRY What kind of association do a car's age and its value usually show?

Wed Mar 8

No association

TRY What kind of association do shoe size and math test score show?

Thu Mar 9

Outliers

TRY Most points lie close to a line, but one point is far away. What is that point called?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Data & Statistics

Lines of Best Fit — **Core:** A good line of fit has about as many points above it as below it.

Mon Mar 13

Predicting

TRY A line of fit for test score vs. hours studied is $y = 5x + 50$. Predict the score for 6 hours.

Tue Mar 14

What the slope means

TRY In $y = 5x + 50$, what does the 5 mean?

Wed Mar 15

What the intercept means

TRY In $y = 5x + 50$, what does the 50 mean?

Thu Mar 16

Writing a line of fit

TRY A line of fit passes through $(2, 30)$ and $(8, 60)$. Write its equation.

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Data & Statistics

Using Linear Models — **Core:** The slope is the rate of change; the intercept is the starting value.

Mon Mar 20
Rate of change
TRY A plant's height is $y = 0.8x + 3$ cm after x days. How fast does it grow?

Tue Mar 21
Starting value
TRY In the same model, how tall was the plant on day 0?

Wed Mar 22
Predicting
TRY Use the model to predict the height on day 20.

Thu Mar 23
Limits of a model
TRY Should you use this model to predict the height on day 1,000? Why or why not?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Data & Statistics

Two-Way Tables — **Core:** Rows and columns show two categories at once.

Mon Apr 3
Reading a table
TRY Of 40 students, 25 play a sport, and 15 of those also play music. How many play a sport but not music?

Tue Apr 4
Completing a table
TRY A class of 60 has 35 girls. 20 girls and 15 boys like soccer. How many boys do not like soccer?

Wed Apr 5
Totals
TRY In the same class, how many students like soccer?

Thu Apr 6
Missing cells
TRY In the same class, how many girls do not like soccer?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Data & Statistics

Relative Frequency & Association — **Core:** Compare percents across rows to look for an association.

Mon Apr 10
Relative frequency
TRY In the class above, 20 of the 35 girls like soccer. About what percent is that?

Tue Apr 11
Comparing rows
TRY 15 of the 25 boys like soccer. What percent is that?

Wed Apr 12
Is there an association?
TRY Is there a strong association between being a boy or girl and liking soccer?

Thu Apr 13
A strong association
TRY 70% of students who sleep 8+ hours eat breakfast, but only 30% of the others do. Is there an association?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Numbers, Equations & Functions — **Core:** Mix the skills from Units 1–3 and show your work.

Mon Apr 17
Review: exponents
TRY Compute $(2^3)^2 \div 2^4$.

Tue Apr 18
Review: scientific notation
TRY Compute $(1.5 \times 10^3)(4 \times 10^2)$.

Wed Apr 19
Review: equations
TRY Solve $3(x + 4) = 5x - 2$.

Thu Apr 20
Timed set: 10 in 20 min
TRY Find the slope of $2y = 6x - 4$.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Test

Core: Practice a full test, then learn from every error.

Mon Apr 24

Review: systems

TRY Solve $x + 2y = 11$ and $x - y = 2$.

Tue Apr 25

Review: Pythagorean theorem

TRY A right triangle has legs 9 and 40. Find the hypotenuse.

Wed Apr 26

Review: volume

TRY Find the volume of a cone with radius 6 and height 5.

Thu Apr 27

Practice Test

Full-length Grade 8 practice test covering the whole year.

Fri Apr 28

Error-log review

Redo every missed problem from the test; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for high school math.

Mon May 1

Light review

TRY Compute $\sqrt{64} + \sqrt[3]{64}$.

Tue May 2

Mixed warm-up set

TRY Write the line with slope $-\frac{1}{2}$ and y-intercept 3.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for high school.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: 3^6
Tue: $5^2 = 25$
Wed: $\frac{1}{8}$ and 1
Thu: $\frac{1}{27}$

Week 2
Mon: 9.3×10^7
Tue: 4.2×10^{-4}
Wed: 6×10^9
Thu: 3,000

Week 3
Mon: 12 and $\frac{7}{8}$
Tue: 5 and -2
Wed: $x = 9$ or $x = -9$
Thu: 7 cm

Week 4
Mon: Rational, irrational, rational, irrational
Tue: $\frac{4}{9}$
Wed: 7 and 8; closer to 7
Thu: 3.1, π, $\sqrt{10}$, $3\frac{1}{4}$

Week 5
Mon: $x = 5$
Tue: $x = 4$
Wed: $x = 17$
Thu: $x = 6$

Week 6
Mon: $x = 5$
Tue: $x = 6$
Wed: $x = 12$
Thu: 6 visits

Week 7
Mon: $x = 5$
Tue: None
Wed: Infinitely many
Thu: 2

Week 8
Mon: 3
Tue: Car B (60 mph)
Wed: -2
Thu: The slope triangles are similar

Week 9
Mon: $m = -3$, $b = 7$
Tue: $y = 2x - 5$
Wed: $y = 3x + 1$
Thu: $y = 25x + 40$

Week 10
Mon: $\frac{1}{16}$
Tue: 4×10^4
Wed: $x = 4$
Thu: 2

Week 11
Mon: No: input 1 has two outputs
Tue: 9
Wed: No
Thu: 8

Week 12
Mon: No
Tue: No; yes
Wed: 2
Thu: It increases quickly, then stays the same

Week 13
Mon: g (rate 4)
Tue: f (2 vs. 1)
Wed: $y = 500 - 20x$
Thu: After 25 minutes

Week 14
Mon: Yes
Tue: $(5, 15)$
Wed: $(7, 3)$
Thu: None (parallel lines)

Week 15
Mon: 15 adult, 15 child
Tue: 200 minutes
Wed: 35 and 21
Thu: 8 nickels, 7 dimes

Week 16
Mon: $(-1, 3)$
Tue: $(5, -2)$
Wed: $(-1, 4)$
Thu: Yes, both

Week 17
Mon: $(6, -9)$
Tue: $(4, 2)$
Wed: 2.5
Thu: Angles

Week 18
Mon: Similar
Tue: A $180°$ rotation about the origin
Wed: 15
Thu: 15

Week 19
Mon: $65°$
Tue: $70°$
Wed: $30°$, $60°$, $90°$
Thu: $115°$

Week 20
Mon: 5 and 6
Tue: $x = 5$
Wed: -4
Thu: $(2, 4)$

Week 21
Mon: Yes (both are 40-60-80)
Tue: 9 m
Wed: 6
Thu: No

Week 22
Mon: 10
Tue: 15
Wed: 12 ft
Thu: $\sqrt{13}$

Week 23
Mon: Yes $(49 + 576 = 625)$
Tue: No $(36 + 49 \neq 81)$
Wed: 40 in.
Thu: 13

Week 24
Mon: 5
Tue: 10
Wed: 12
Thu: 13

Week 25
Mon: 90π
Tue: 30π
Wed: 4
Thu: 3

Week 26
Mon: 36π
Tue: 144π
Wed: $1{,}000 - \frac{500\pi}{3}$
Thu: $\frac{40\pi}{3}$

Week 27
Mon: Positive
Tue: Negative
Wed: None
Thu: An outlier

Week 28
Mon: 80
Tue: Each extra hour adds about 5 points
Wed: The predicted score with 0 hours of study
Thu: $y = 5x + 20$

Week 29
Mon: 0.8 cm per day
Tue: 3 cm
Wed: 19 cm
Thu: No: real plants stop growing

Week 30
Mon: 10
Tue: 10
Wed: 35
Thu: 15

Week 31
Mon: About 57%
Tue: 60%
Wed: No: 57% and 60% are close
Thu: Yes

Week 32
Mon: 4
Tue: 6×10^5
Wed: $x = 7$
Thu: 3

Week 33
Mon: $(5, 3)$
Tue: 41
Wed: 60π

Week 34
Mon: 12
Tue: $y = -\frac{1}{2}x + 3$

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